

Session 1
Start of the Lesson (10 minutes)
Answer the questions below without checking your notes. Then, share your answers with your group. As a group, discuss your ideas and agree on one best insight to record and share with the class. 1. Why is a Work Breakdown Structure (WBS) important before starting the prototype build? <u>because it helps divide the project into smaller and manageable tasks. It allows the team to organize responsibilities, monitor progress, and complete the prototype efficiently within the timeline.</u> 2. What was the most important criterion your team used when selecting your final solution idea? <u>the ability of the system to solve real problems related to misinformation and improper use of dietary supplements while also being practical and useful for users.</u> 3. How does a Gantt Chart help your team manage time and prepare for prototyping? <u>helps the team schedule tasks, assign deadlines, track progress, and ensure that all members complete their responsibilities on time during the prototype phase.</u> 4. Looking back at Design Thinking, how does the Prototype Phase connect to what you've done in Ideation and Definition? (<i>Hint: Think about how ideas turn into tangible models.</i>) <u>The Prototype Phase transforms the ideas and identified problems from the Ideation and Definition stages into an actual model or system design that can be tested and improved.</u>

MAIN TASK

Try It Out: What Makes a Tool the “Right” Tool?

1.) What kinds of tools or technologies do you already know how to use?

Our team already knows how to use Figma for design, React.js for frontend development, Node.js and Express.js for backend development, MongoDB for database management, and GitHub for collaboration.

2.) What new tools might help your team build a better prototype?

Artificial Intelligence APIs, chatbot integration tools, and UI design libraries may help improve the functionality and appearance of our prototype.

3.) What challenges could happen if the team chooses tools that are too advanced or don't fit the project goals?

The team may experience delays, technical difficulties, compatibility issues, and confusion if the tools are too complex or not suitable for the project requirements.

Let's Practice: Build Your Prototype Foundation (50 minutes)**Activity 1: Pick the Best Tools for the Job**

Objective: Choose the tools and technologies your group will use to build your prototype.

Instructions:

1. Review your project goals and chosen solution.
2. Discuss which tools best match your system's requirements and your team's skills.
3. Assign responsibilities — who will handle which tool or component?
4. Record your decisions in the Tool Stack Table below.

Tool Stack Table

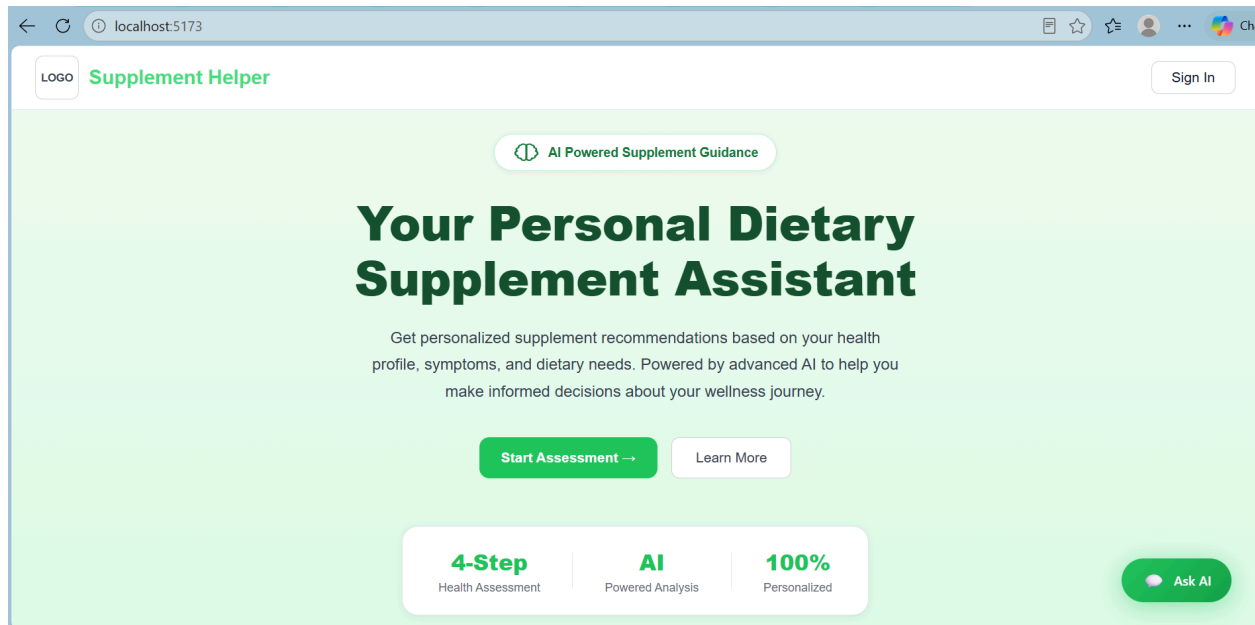
Category	Tool/Technology	Purpose	Assigned To
Design	Figma	Create wireframes and UI designs	UI/UX Designer
Frontend	React.js	Develop the user interface and web pages	Frontend Developer
Backend	Node.js, Express.js	Handle server-side logic and API integration	Backend Developer
Database	MongoDB	Store user and supplement information	Database Manager
Others	GitHub, VS Code	Collaboration, version control, and coding	Whole Team

Activity 2: Create Our First Prototype

Objective: Build a first version of your solution as a low- or mid-fidelity prototype.

Instructions:

1. Divide your team into smaller groups or pairs.
Example: Pair A = UI/Design; Pair B = Backend/Functionality.
2. Use your chosen tools to build a simple prototype that focuses on the core features of your system.
 - **Low-fidelity:** Sketches, wireframes, mockups (e.g., Canva, Figma).
 - **Mid-fidelity:** Clickable model or basic coded demo (e.g., HTML/CSS, app screens).
3. Save **screenshots, wireframes, or demo links** as documentation.
4. Write short notes describing:
 - What you built
 - What tools you used
 - Any challenges or issues encountered



WHAT WE BUILT:

We created a low- to mid-fidelity prototype of the SuppleWise web application that includes the homepage, login page, and supplement recommendation page.

WHAT TOOLS WE USED:

We used Figma for the interface design, React.js for frontend development, Node.js and Express.js for backend functions, MongoDB for data storage, and GitHub for collaboration.

CHALLENGES OR ISSUES ENCOUNTERED:

One challenge encountered was integrating frontend and backend components properly. The team solved this issue by reviewing documentation, testing APIs, and collaborating on debugging tasks.

Session 2
Start of the Lesson (10 minutes)
Answer the questions below. 1. What tools or technologies did your team choose for your design, frontend, backend, and database, and why were these the best fit for your project? Our team chose Figma for design because it is easy to use for creating wireframes and prototypes. React.js was selected for frontend development because it allows responsive and interactive user interfaces. Node.js and Express.js were used for backend development because they efficiently handle server-side functions and APIs. MongoDB was chosen as the database because it is flexible and suitable for storing user and supplement data. 2. What part of your prototype (UI, feature, or function) was successfully built or visualized during the last session? The login page, homepage layout, and supplement recommendation interface were successfully designed. 3. What was the biggest challenge your team faced while building the first version of your prototype, and how did you try to solve it? The biggest challenge was connecting the frontend interface with backend functions. The team solved this problem through collaboration, debugging, and researching proper API integration methods. 4. How did creating a low- or mid-fidelity prototype help your team test or communicate your idea more effectively? Creating a prototype helped the team visualize the system clearly, identify design issues early, and communicate the project idea more effectively to advisers and group members.

Main Task - Try It Out: How Did We Do?

1. What parts of your prototype are already working or visible?

The user interface, login system design, homepage, and supplement recommendation layout are already visible and partially functional.

2. What features are still missing or need improvement?

The AI recommendation system, supplement verification feature, and database integration still need improvement and further testing.

3. How close are you to your goals and timeline from Module 5?

Our team is progressing according to the planned timeline, although some backend and AI-related features still require additional development.

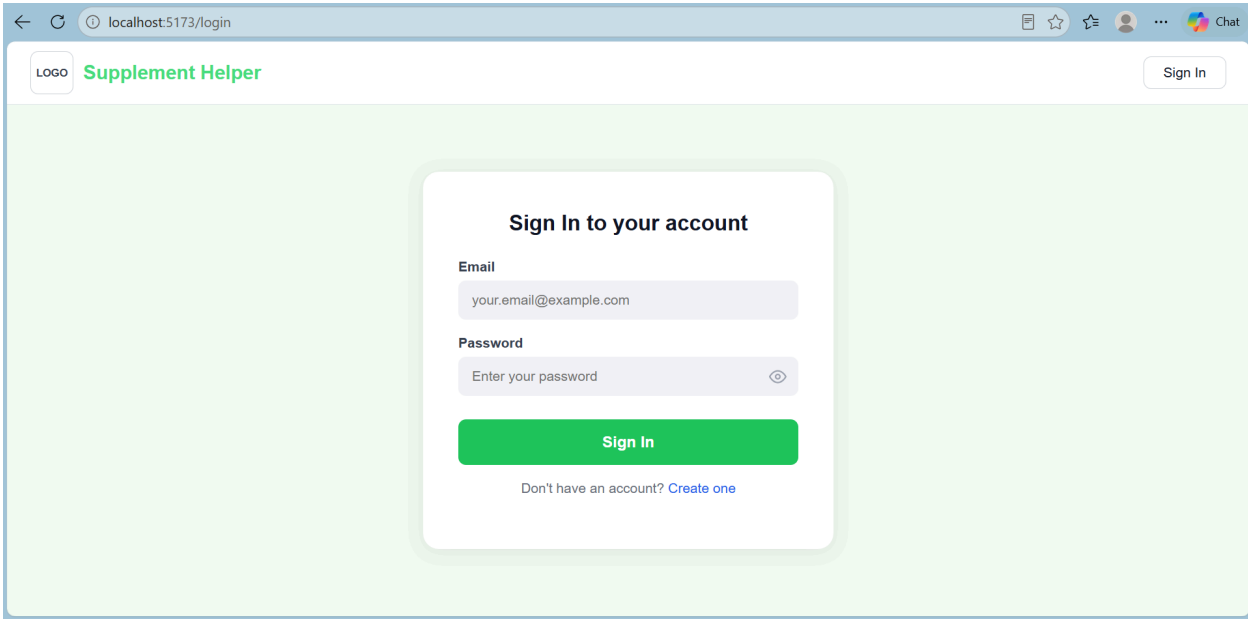
Let's Practice: Activity 3 - Show and Record Our Prototype (40 minutes)

Feature/Part Built	Screenshot/Diagram	Notes/Challenges	Link to WBS/Gantt

Login and Registration Page		Needed improvements in form validation and responsiveness	Yes – completed according to schedule
Homepage Interface		Minor layout adjustments were needed	Yes – aligned with planned tasks
Supplement Recommendation		AI recommendation feature still under development	Yes – currently in progress

SCREENSHOT:

Login and Registration Page



←localhost:5173/signup

Sign In

LOGO

Supplement Helper

Create Account

Name

Enter your Name

Email

Your.Email@example.com

Password

Enter your password

Confirm Password

Confirm your password

Sign In

Already Have an Account? [Login](#)

HomePage

←localhost:5173

Sign In

LOGO

Supplement Helper

AI Powered Supplement Guidance

Your Personal Dietary Supplement Assistant

Get personalized supplement recommendations based on your health profile, symptoms, and dietary needs. Powered by advanced AI to help you make informed decisions about your wellness journey.

Start Assessment →

Learn More

4-Step

Health Assessment

AI

Powered Analysis

100%

Personalized

Ask AI

Supplement Recommendation

Magnesium Glycinate

High

Very Poor sleep quality detected. Magnesium glycinate activates GABA receptors to quiet the nervous system. It is the most clinically studied supplement for sleep quality improvement.

Dosage: 400mg daily **Timing:** 30-60 minutes before bed

Melatonin (low dose)

Medium

Very Poor sleep quality reported. Low-dose melatonin helps reset the circadian rhythm and reduce sleep onset time. Lower doses (0.5-1mg) are more physiologically appropriate than high doses.

Dosage: 0.5â€“1mg **Timing:** 30 minutes before target bedtime

⚠ Avoid with sedatives or alcohol

Think Back and Think About (10 minutes)

Think Back: How did choosing the right tools and technologies in Session 1 help your team build and improve your prototype more efficiently in Session 2?

Choosing the right tools helped our team work faster and build the prototype more efficiently. The tools made designing, coding, and organizing tasks easier during development.

Think About: What did you learn about **teamwork, problem-solving, and documentation** while creating and recording your prototype? How will these lessons help you as you continue developing your Capstone Project?

We learned the importance of teamwork, communication, problem-solving, and proper documentation. These lessons will help us manage tasks better, solve issues faster, and improve our Capstone Project development.